

Problem Statement

- Group travel planning is challenging because **each traveller has different interests, preferences, budgets, and personal requirements**.
- Majority voting alone may cause **minority preferences to be overlooked**, resulting in an itinerary that does not satisfy everyone.
- Important individual requirements, such as **food allergies or dietary restrictions**, may not be considered when choosing activities or places.
- Travellers need to manually compare everyone's preferences, vote on activities, and arrange them into a suitable schedule, making the process **time-consuming and complicated**.
- Existing travel planners often generate itineraries without sufficiently considering **the collective preferences and individual priorities of a group**.

Proposed Solution

- Develop an **AI-powered group travel planner** that creates a balanced itinerary based on the needs and preferences of all travellers.
- Allow each traveller to provide:
 - **Personal preferences and interests**
 - **Allergies / dietary restrictions**
 -  **Must-do activity**
 - Budget preferences
- Provide a **voting system** where group members vote on suggested activities.
- Use voting results to identify **popular activities**, while ensuring the itinerary does not rely solely on majority votes.
- Treat **must-do activities as individual priorities**, with the system attempting to include at least one important activity for each traveller.
- Treat **allergies and other requirements as constraints** when recommending activities, restaurants, and places.
- Use AI to automatically generate an itinerary by balancing:
 -  Group voting results
 -  Individual must-do activities
 -  Personal preferences
 -  Priority requirements
 -  Budget
 -  Available time
 -  Location and travel distance
- Validate the generated itinerary for **time conflicts, budget limitations, and requirement conflicts**.
- If all preferences cannot realistically be accommodated, the system **suggests alternatives or compromises**.

- Overall, the solution aims to create a trip that **does not simply satisfy the majority, but balances the preferences and priorities of the whole group.**

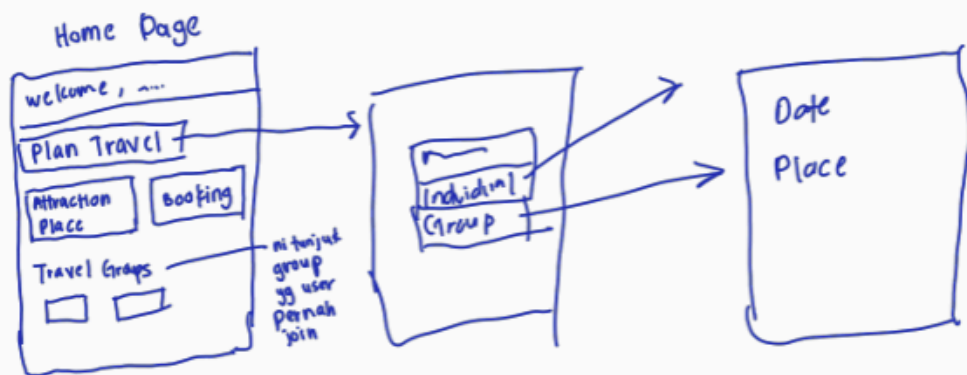
Flow

1. Create Account / Login
2. Create a New Trip
3. Enter Trip Details
 - Destination
 - Travel dates / duration
 - Budget
 - Number of travellers
4. Invite / Join Group
5. Each Traveller Enters Personal Information
 - Interests / preferences
 - Food preferences
 - Allergies / dietary restrictions
 - Must-do activity
6. System Collects and Combines Group Preferences
7. AI Suggests Suitable Activities / Places
8. Group Members Vote on Suggested Activities
9. System Analyses the Results
 - Voting results
 - Must-do activities
 - Personal preferences
 - Allergies / restrictions
 - Budget
 - Available time
10. AI Generates a Group Itinerary
11. System Checks the Itinerary
 - Requirements are considered
 - Must-do activities are included where feasible
 - Within budget
 - No time conflicts
 - Reasonable travel distance
12. Display Generated Itinerary
13. Group Reviews and Adjusts the Itinerary
14. AI Regenerates / Suggests Alternatives if Needed
15. Group Approves Final Itinerary
16. Final Itinerary Ready for the Trip

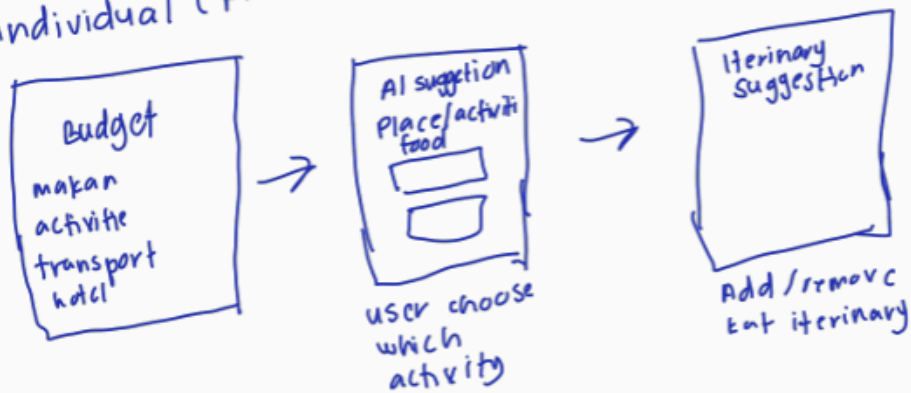


Login/Register

username
password



Individual (plan travel)



Group

Budget

min max

voting budget

Average Budget: ~

Place Suggestion

[]

[]

- Shows majority budget & suggest place / activity / food
- user choose which one

Itinerary

[]

[]

[]

- AI suggest itinerary
- add / remove

Place Attraction

Browse Activity / Place

Country / State

category

↳ nature

↳ adventure

Nature

[]

[]

[]

About Attraction

Place

Explanation

~

~

Booking

Hotel

Activity Ticket

Hotel website
(booking.com / airbnb)

Place
(activity)